

Answer **all** questions.

1. Different theories have been proposed to explain the evolution of the three Domains of life. **Image 1.1** and **Image 1.2** show different phylogenetic trees to explain the relationships between the three Domains.

Image 1.1

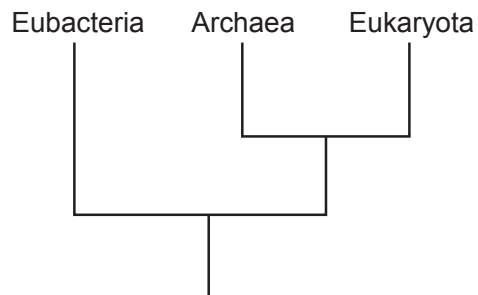
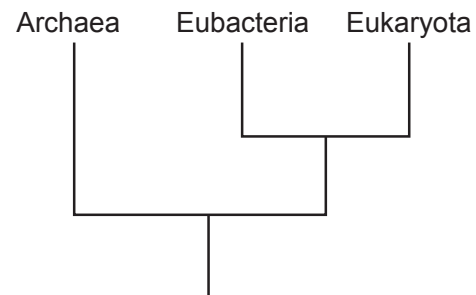


Image 1.2



- (a) The last universal common ancestor (LUCA) is the most recent population of organisms from which all organisms now living on Earth have evolved.

Place a cross (x) on **Image 1.1** to show the position of LUCA.

[1]

- (b) State **two** features found in the cells of organisms in the Domain Eukaryota that are not found in the cells of organisms in the Domains Archaea and Eubacteria.

[2]

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- (c) Analysis of the nucleotide sequences at specific positions (numbered 1–8) in molecules of ribosomal RNA (rRNA) have been used to provide evidence for the evolution of the three Domains. This is shown in **Table 1.3**.

Table 1.3

Position	Nucleotide sequence	Occurrence in species analysed / %		
		Archaea	Eubacteria	Eukaryota
1	CACUUG	0	>95	0
2	AAACUCAA	3	100	0
3	AAACUAAAG	100	0	100
4	CUCAAUG	100	<1	100
5	CAACCUUCG	0	>95	0
6	UCCCUG	>95	0	100
7	UACACACCG	0	>99	100
8	CACACACCG	100	0	0



- (i) It has been concluded that **Image 1.1** provides the most likely explanation for the evolution of the three Domains.

Use **Table 1.3** to identify **one** piece of evidence that supports this decision and **one** piece of evidence that does not support this decision. Explain your answers. [2]

One piece of evidence that supports the decision:

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One piece of evidence that does not support the decision:

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- (ii) In this analysis, the rRNA molecules from each domain were extracted from large numbers of different species. Explain why this was necessary. [1]

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- (iii) Explain why evidence obtained from biochemical analysis is of more use in assessing the relatedness of organisms than the comparison of morphological features. [1]

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